

Q4-B: Find All Ideals in the Ring $\mathbb{Z}_{18}$. Generalize to $\mathbb{Z}_n$

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Preliminaries

Definition 1 (Ring and Ideal). A **ring** $(R, +, \cdot)$ is a set with two binary operations satisfying the ring axioms. A non-empty subset $I \subseteq R$ is an **ideal** of R if:

- (i) $(I, +)$ is a subgroup of $(R, +)$, and
- (ii) For all $r \in R$ and $a \in I$: $ra \in I$ and $ar \in I$ (absorption property).

Definition 2 (Principal Ideal). For a commutative ring R with unity and any element $a \in R$, the **principal ideal generated by a** is

$$\langle a \rangle = aR = \{ ar \mid r \in R \}.$$

Theorem 1. $\mathbb{Z}_n$ is a commutative ring with unity. Every ideal of $\mathbb{Z}_n$ is a principal ideal. Specifically, the ideals of $\mathbb{Z}_n$ are exactly the subgroups of $(\mathbb{Z}_n, +)$, which are in bijection with the divisors of n .

Proof sketch. Since $\mathbb{Z}_n$ is a principal ideal ring (a quotient of the PID $\mathbb{Z}$), every ideal of $\mathbb{Z}_n$ is principal. Moreover, every additive subgroup of $\mathbb{Z}_n$ is automatically an ideal, since for any $a \in I \subseteq \mathbb{Z}_n$ and $r \in \mathbb{Z}_n$, the product ra is a repeated sum of a , hence lies in I . The additive subgroups of $\mathbb{Z}_n$ correspond bijectively to divisors d of n via $d \mapsto \langle d \rangle$. $\square$

Part 1: All Ideals of $\mathbb{Z}_{18}$

The ring $\mathbb{Z}_{18} = \{0, 1, 2, \dots, 17\}$ has order 18. We first find all divisors of 18:

$$18 = 2 \cdot 3^2, \quad \text{divisors of 18 : } 1, 2, 3, 6, 9, 18.$$

Each divisor d of 18 gives a unique ideal $\langle d \rangle$ of $\mathbb{Z}_{18}$.

For each divisor $d \mid 18$, the ideal $\langle d \rangle$ consists of all multiples of d modulo 18, and has order $18/d$:

1. $d = 1$:

$$\langle 1 \rangle = \mathbb{Z}_{18} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17\}, \quad |\langle 1 \rangle| = 18.$$

This is the **improper (whole ring) ideal**.

2. $d = 2$:

$$\langle 2 \rangle = \{0, 2, 4, 6, 8, 10, 12, 14, 16\}, \quad |\langle 2 \rangle| = 9.$$

This is the ideal of all even residues modulo 18.

3. $d = 3$:

$$\langle 3 \rangle = \{0, 3, 6, 9, 12, 15\}, \quad |\langle 3 \rangle| = 6.$$

4. $d = 6$:

$$\langle 6 \rangle = \{0, 6, 12\}, \quad |\langle 6 \rangle| = 3.$$

5. $d = 9$:

$$\langle 9 \rangle = \{0, 9\}, \quad |\langle 9 \rangle| = 2.$$

6. $d = 18$:

$$\langle 18 \rangle = \langle 0 \rangle = \{0\}, \quad |\langle 18 \rangle| = 1.$$

This is the **zero ideal** (trivial ideal).

Hence $\mathbb{Z}_{18}$ has exactly **six ideals**:

Ideal	Generator d	Elements	Order
$\langle 1 \rangle$	1	$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17\}$	18
$\langle 2 \rangle$	2	$\{0, 2, 4, 6, 8, 10, 12, 14, 16\}$	9
$\langle 3 \rangle$	3	$\{0, 3, 6, 9, 12, 15\}$	6
$\langle 6 \rangle$	6	$\{0, 6, 12\}$	3
$\langle 9 \rangle$	9	$\{0, 9\}$	2
$\langle 0 \rangle$	0	$\{0\}$	1

Remark 1. There is a natural **inclusion-reversing** (containment) relationship among the ideals, reflecting the divisibility lattice of 18:

$$\langle 0 \rangle \subset \langle 9 \rangle \subset \langle 3 \rangle \subset \langle 1 \rangle = \mathbb{Z}_{18},$$

$$\langle 0 \rangle \subset \langle 6 \rangle \subset \langle 2 \rangle \subset \langle 1 \rangle = \mathbb{Z}_{18},$$

$$\langle 0 \rangle \subset \langle 6 \rangle \subset \langle 3 \rangle \subset \langle 1 \rangle = \mathbb{Z}_{18}.$$

In general, $\langle d_1 \rangle \subseteq \langle d_2 \rangle$ if and only if $d_2 \mid d_1$.

Part 2: Generalization to $\mathbb{Z}_n$

Theorem 2 (Ideals of $\mathbb{Z}_n$). *The ideals of $\mathbb{Z}_n$ are in bijection with the positive divisors of n . Explicitly:*

(i) *For every divisor $d \mid n$, the set*

$$\langle d \rangle = \{0, d, 2d, 3d, \dots, n - d\}$$

is an ideal of $\mathbb{Z}_n$, with $|\langle d \rangle| = n/d$.

(ii) Every ideal of $\mathbb{Z}_n$ is of this form.

(iii) There are exactly $\tau(n)$ ideals in $\mathbb{Z}_n$, where $\tau(n)$ denotes the number of positive divisors of n .

Proof. (i) $\langle d \rangle$ is an ideal.

Let $d \mid n$. The set $\langle d \rangle = d\mathbb{Z}_n = \{0, d, 2d, \dots, (n/d - 1)d\}$ is a cyclic subgroup of $(\mathbb{Z}_n, +)$ of order n/d . To verify the absorption property: for any $r \in \mathbb{Z}_n$ and $a = kd \in \langle d \rangle$,

$$r \cdot a = r \cdot kd = (rk)d \equiv (rk \bmod n/d) \cdot d \pmod{n},$$

which is again a multiple of d , hence in $\langle d \rangle$. Thus $\langle d \rangle$ is an ideal.

(ii) Every ideal of $\mathbb{Z}_n$ has this form.

Let I be any ideal of $\mathbb{Z}_n$. Then I is in particular an additive subgroup of $\mathbb{Z}_n$. By the classification of subgroups of cyclic groups, every subgroup of $\mathbb{Z}_n$ is cyclic, generated by some divisor d of n :

$$I = \langle d \rangle \quad \text{for some } d \mid n.$$

Since every additive subgroup of $\mathbb{Z}_n$ is automatically an ideal (as $\mathbb{Z}_n$ is commutative and multiplication by any r sends multiples of d to multiples of d), the ideals and additive subgroups of $\mathbb{Z}_n$ coincide.

(iii) Count.

The divisors of n are in bijection with the ideals of $\mathbb{Z}_n$, so the number of ideals equals $\tau(n)$, the number of positive divisors of n . $\square$

Corollary 1 (Inclusion of Ideals). For divisors d_1, d_2 of n :

$$\langle d_1 \rangle \subseteq \langle d_2 \rangle \iff d_2 \mid d_1.$$

In particular:

- The largest ideal is $\langle 1 \rangle = \mathbb{Z}_n$ (the whole ring).
- The smallest ideal is $\langle n \rangle = \langle 0 \rangle = \{0\}$ (the zero ideal).

Corollary 2 (Maximal and Prime Ideals of $\mathbb{Z}_n$). An ideal $\langle d \rangle$ of $\mathbb{Z}_n$ is:

- **Maximal** if and only if $d = p$ is prime and $p \mid n$. In this case $\mathbb{Z}_n/\langle p \rangle \cong \mathbb{Z}_p$ is a field.
- **Prime** if and only if $d = p$ is prime and $p \mid n$.

(In $\mathbb{Z}_n$, the maximal ideals and prime ideals coincide.)

Example 1 (Verification for $\mathbb{Z}_{18}$). We have $n = 18 = 2 \cdot 3^2$, so $\tau(18) = (1+1)(2+1) = 6$, confirming the six ideals found in Part 1. The maximal (and prime) ideals are $\langle 2 \rangle$ and $\langle 3 \rangle$, corresponding to the prime divisors 2 and 3 of 18.

Theorem 3 (General Formula for Ideals of $\mathbb{Z}_n$). If $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$ is the prime factorisation of n , then the number of ideals of $\mathbb{Z}_n$ is

$$\tau(n) = (a_1 + 1)(a_2 + 1) \cdots (a_k + 1).$$

Each ideal is of the form $\langle d \rangle$ for a unique divisor d of n , and consists of all multiples of d modulo n :

$$\langle d \rangle = \{0, d, 2d, \dots, (n/d - 1) \cdot d\}, \quad |\langle d \rangle| = \frac{n}{d}.$$

Summary

Ring	Number of Ideals	Ideals
$\mathbb{Z}_{18}$	$\tau(18) = 6$	$\langle 1 \rangle, \langle 2 \rangle, \langle 3 \rangle, \langle 6 \rangle, \langle 9 \rangle, \langle 0 \rangle$
$\mathbb{Z}_n$, general	$\tau(n)$	$\langle d \rangle$ for each $d \mid n$
$\mathbb{Z}_p$, p prime	2	$\langle 0 \rangle$ and $\langle 1 \rangle = \mathbb{Z}_p$ only

The last row reflects the well-known fact that $\mathbb{Z}_p$ is a **field** (hence a simple ring) whenever p is prime, having no proper non-trivial ideals. ■